Ratios, Proportions & Percentages

ClipSAT Printable Practice Suite (clipsat.org) • Topic 2.01

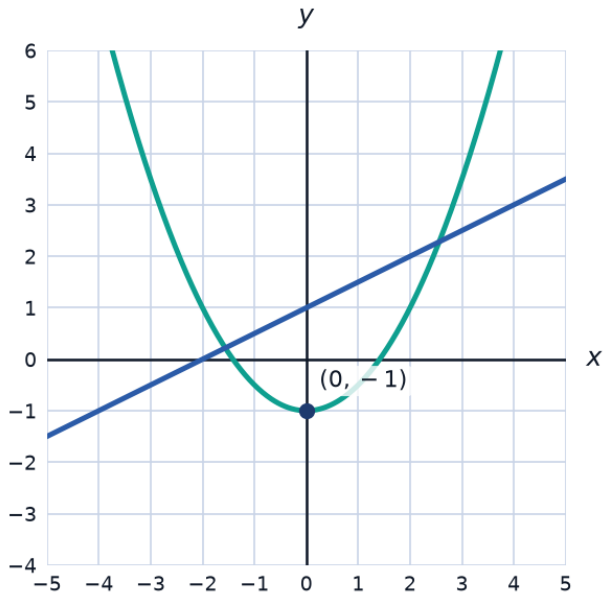
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Simplifying ratios

Simplify the ratio $18 : 24$.

2 [Foundational] • Target: Simplifying ratios

Simplify the ratio $45 : 60 : 75$.



3 [Foundational] • Target: Simplifying ratios

Simplify the ratio $36 : 8$.

4 [Foundational] • Target: Simplifying ratios

Express "12 boys to 18 girls" as a ratio in simplest form.

5 [Foundational] • Target: Sharing a quantity in a given ratio

Share 150 in the ratio $2 : 3$.

6 [Foundational] • Target: Sharing a quantity in a given ratio

Share 220 among three people in the ratio $3 : 4 : 4$.

7 [Foundational] • Target: Sharing a quantity in a given ratio

A prize of 900 riyals is shared between Sara and Layla in the ratio $5 : 4$. Find each share.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Sharing a quantity in a given ratio

Divide 84 kg of flour in the ratio $1:2:4$.

9 [Intermediate] • Target: Equivalent ratios and proportions

If $3:5 = x:20$, find x.

10 [Intermediate] • Target: Equivalent ratios and proportions

If $a:7 = 12:21$, find a.

11 [Intermediate] • Target: Equivalent ratios and proportions

A map has scale $1:50{,}000$. A distance of 3 cm on the map represents how many meters in reality?

12 [Intermediate] • Target: Equivalent ratios and proportions

Solve for y: $\frac{4}{9} = \frac{y}{63}$.

13 [Intermediate] • Target: Percent of a number

Find 35% of 240.

14 [Intermediate] • Target: Percent of a number

Find 12% of 150.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Percent of a number • Cross-Lesson: 5.01 Quantitative Comparisons

What is 8% of 650?

16 [Challenging] • Target: Percent of a number • Cross-Lesson: 6.01 Sequences & Patterns

Find 150% of 40.

17 [Challenging] • Target: Converting between fractions, decimals and percentages • Cross-Lesson: 7.01 Data Analysis & Probability

Write $\frac{3}{8}$ as a percent.

18 [Challenging] • Target: Converting between fractions, decimals and percentages • Cross-Lesson: 8.01 Word Problems & Reasoning

Write 0.45 as a fraction in lowest terms.

19 [Challenging] • Target: Converting between fractions, decimals and percentages • Cross-Lesson: 1.01 Arithmetic & Number Sense

Write 62.5% as a fraction in lowest terms.

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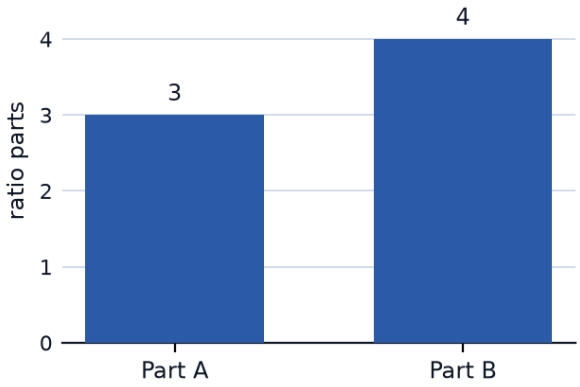
20 [Challenging] • Target: Converting between fractions, decimals and percentages • Cross-Lesson: 3.01 Algebra Essentials

Write $\frac{7}{40}$ as a decimal.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: MCQ — Sharing in a given ratio

The bar chart shows the ratio in which a total of 84 is to be shared. Find the larger share.



a. 36

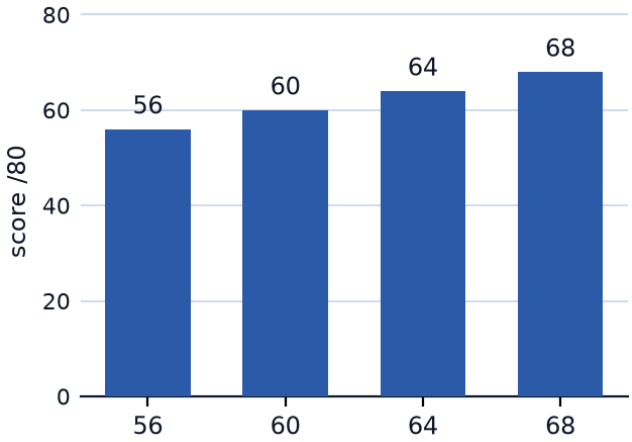
b. 21

c. 48

d. 63

22 [Foundational] • Target: MCQ — Percent of a number

The bar chart shows four students' quiz scores out of 80. Which score is exactly 75%?



a. 68

b. 64

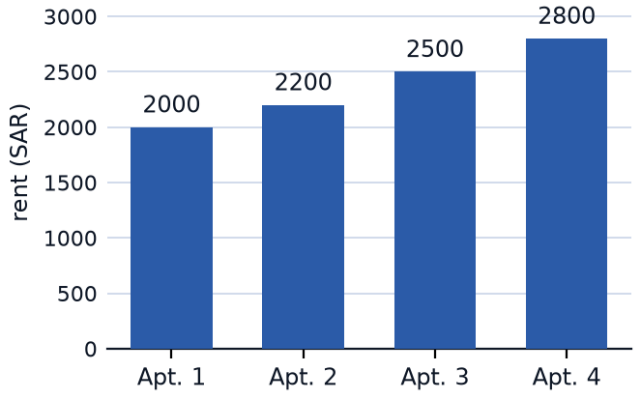
c. 60

d. 56

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23 [Foundational] • Target: MCQ — Percent of a number

The bar chart shows monthly rent for four apartments. 8% of which apartment's rent equals 200 SAR?



a. Apt. 2

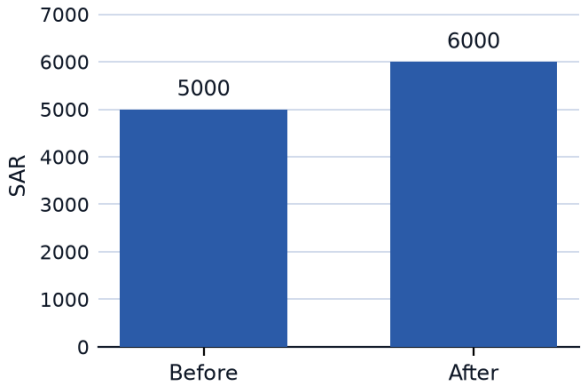
b. Apt. 1

c. Apt. 3

d. Apt. 4

24 [Foundational] • Target: MCQ — Percentage increase & decrease

The bar chart shows a salary before and after a raise. What percentage increase does it represent?



a. 15%

b. 500%

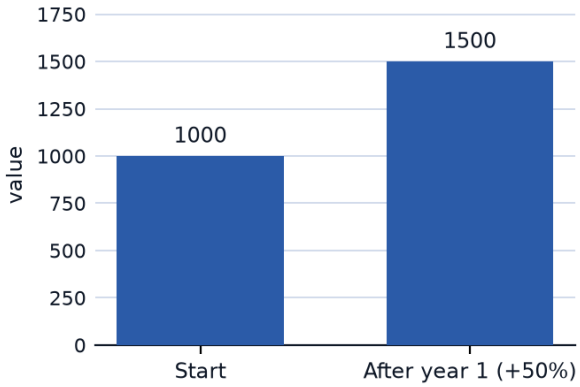
c. 20%

d. 25%

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25 [Foundational] • Target: MCQ — Successive percentage changes

The bar chart shows a stock's value at the start and after rising 50% in year one. It then falls 50% in year two. Find its final value.



a. 500

b. 900

c. 1,000

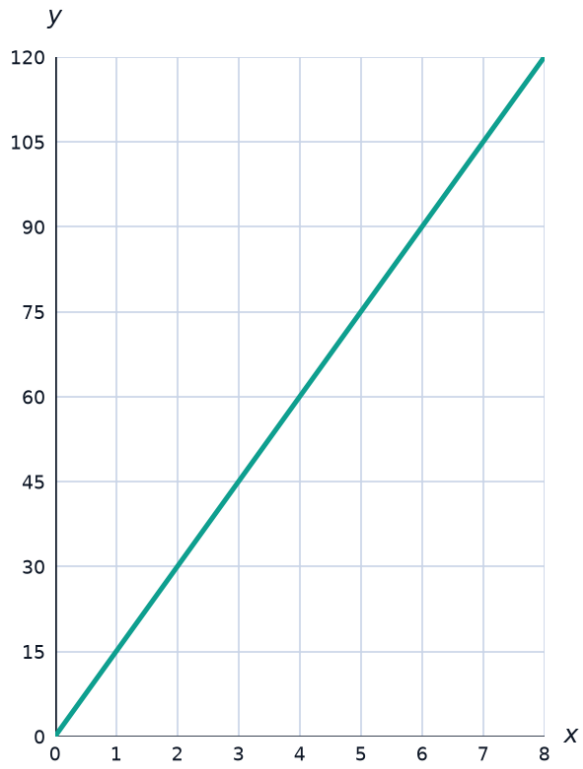
d. 750

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26 [Foundational] • Target: MCQ — Direct proportion

The graph shows the direct proportion between the number of hours x worked and pay y in SAR, $y = 15x$.

How much is earned for 6 hours?



a. 105

b. 90

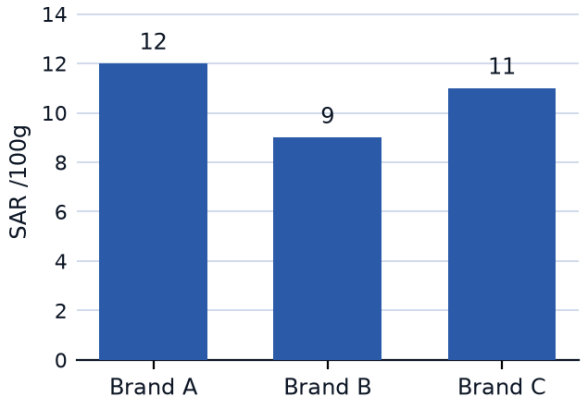
c. 75

d. 21

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27 [Foundational] • Target: MCQ — Best value & unit price

The bar chart shows the price per 100 g for three brands of coffee. Which brand offers the best value?



a. Brand B

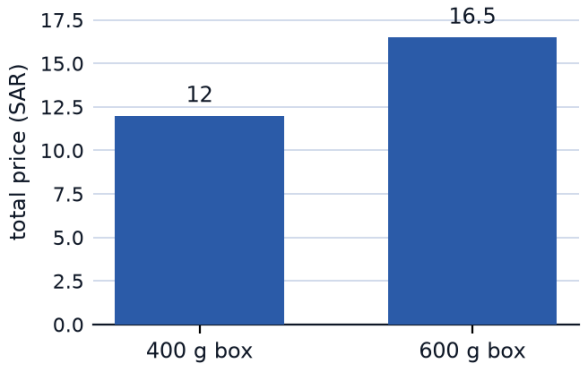
b. All are equal

c. Brand C

d. Brand A

28 [Foundational] • Target: MCQ — Best value & unit price

The bar chart shows the total price of two cereal boxes of different sizes: 400 g and 600 g. Which has the lower price per 100 g?



a. Cannot be determined

b. They are equal

c. The 600 g box

d. The 400 g box

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: MCQ — Simplifying & writing ratios

Simplify the ratio $42 : 56$.

a. $3 : 4$

b. $6 : 8$

c. $7 : 8$

d. $3 : 8$

30 [Intermediate] • Target: MCQ — Sharing in a given ratio

Two business partners share a profit of $18{,}000$ SAR in the ratio $5 : 7$. Find the larger share.

a. $10{,}500$ SAR

b. $9{,}000$ SAR

c. $7{,}500$ SAR

d. $12{,}857$ SAR

31 [Intermediate] • Target: MCQ — Equivalent ratios & scale

Which ratio is equivalent to $4 : 9$?

a. $9 : 4$

b. $20 : 45$

c. $8 : 16$

d. $16 : 37$

32 [Intermediate] • Target: MCQ — Fraction, decimal & percent conversion

Write $\frac{7}{20}$ as a percent.

a. 35%

b. 3.5%

c. 7%

d. 70%

33 [Intermediate] • Target: MCQ — Percentage increase & decrease

A shirt costs 120 SAR after a 20% discount. Find the original price.

a. 150 SAR

b. 144 SAR

c. 100 SAR

d. 140 SAR

34 [Intermediate] • Target: MCQ — Finding the original value

After including a 15% service charge, a restaurant bill is 138 SAR. Find the bill before the charge.

a. 158.70 SAR

b. 120 SAR

c. 123 SAR

d. 117.30 SAR

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35 [Intermediate] • Target: MCQ — Direct proportion

A car uses 8 liters of fuel to travel 96 km. At the same rate, how far can it travel on 20 liters?

a. 200

b. 260

c. 240

d. 180

36 [Intermediate] • Target: MCQ — Best value & unit price

A 2 L bottle of juice costs 18 SAR. Find the price per 100 mL.

a. 9 SAR

b. 0.18 SAR

c. 1.80 SAR

d. 0.90 SAR

37 [Intermediate] • Target: MCQ — Fraction, decimal & percent conversion

Find 15% of 160.

a. 19

b. 29

c. 36

d. 24

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: MCQ — Percentage increase & decrease • Cross-Lesson: 8.01 Word Problems & Reasoning

Find 10% of 400.

a. 52

b. 40

c. 35

d. 45

39 [Challenging] • Target: MCQ — Finding the original value • Cross-Lesson: 1.01 Arithmetic & Number Sense

Find 5% of 100.

a. 5

b. 17

c. 10

d. 25

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40 [Challenging] • Target: MCQ — Successive percentage changes • Cross-Lesson: 3.01 Algebra Essentials

Decrease 210 by 10%.

a. 185

b. 189

c. 198

d. 193

41 [Challenging] • Target: MCQ — Direct proportion • Cross-Lesson: 4.01 Geometry & Measurement

Increase 310 by 10%.

a. 350

b. 345

c. 341

d. 337

42 [Challenging] • Target: MCQ — Inverse proportion • Cross-Lesson: 5.01 Quantitative Comparisons

Decrease 340 by 50%.

a. 166

b. 179

c. 174

d. 170

43 [Challenging] • Target: MCQ — Best value & unit price • Cross-Lesson: 6.01 Sequences & Patterns

Share 390 among three people in the ratio $7:3:3$.

a. $210, 90, 90$

b. $90, 90, 210$

c. $240, 90, 90$

d. $210, 90, 60$

44 [Challenging] • Target: MCQ — Word problems • Cross-Lesson: 7.01 Data Analysis & Probability

7 liters of paint cost 98 riyals. Find the cost of 5 liters of paint at the same rate.

a. 80

b. 56

c. 70

d. 84

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 8.01 Word Problems & Reasoning

After a 50% increase, a quantity is now 90. Find the original value.

a. 60

b. 70

c. 50

d. 90

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